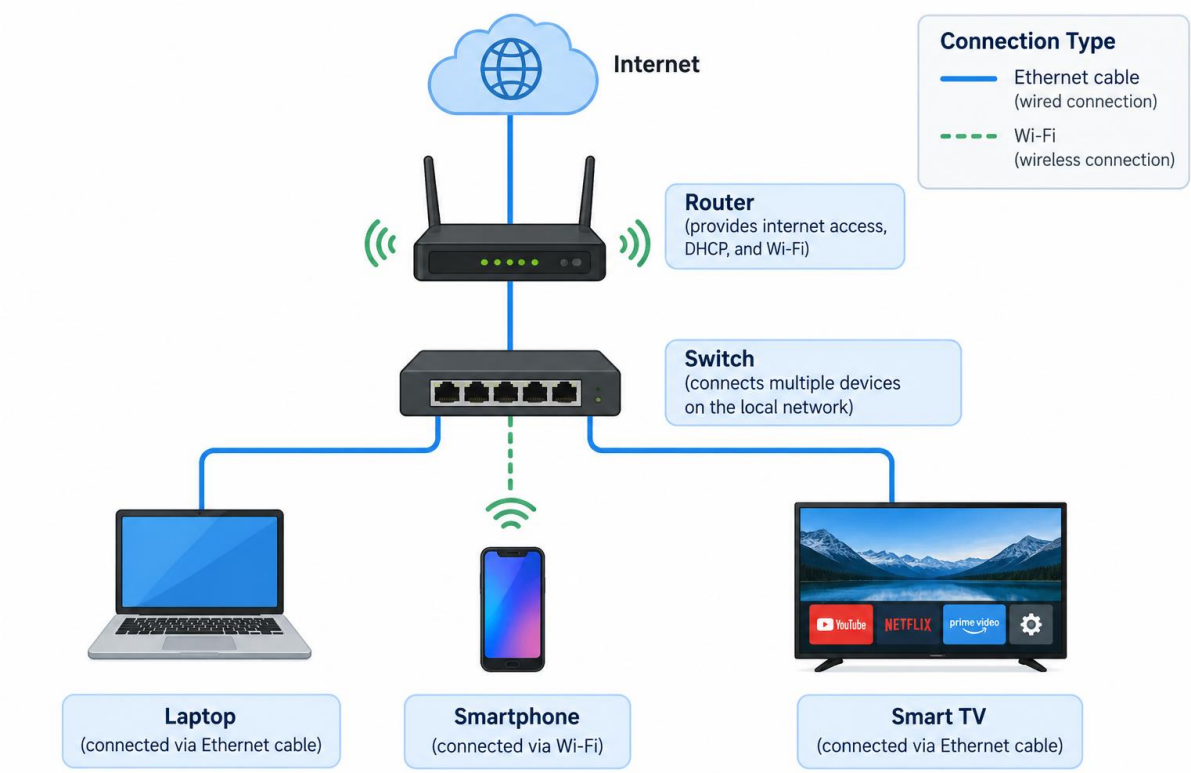
Network Devices

Task 1:- Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.



Priyansh Bhaliya

Task 2:- Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.).

Network device	Main function	Real-world use case	Example app or service
1. Router	Connects different networks and directs data between them.	Connects a home network to the internet so multiple devices can access online services.	YouTube streaming, WhatsApp, Google
2. Switch	Connects multiple devices within a local area network (LAN).	Connects computers and gaming consoles in a college lab or gaming zone for fast local communication.	LAN gaming, file sharing, online gaming
3. Hub	Broadcasts incoming data to all connected ports.	Can connect several computers in a small legacy network for basic communication.	Basic file sharing between older computers
4. Wireless Access Point (AP)	Provides Wi-Fi connectivity to wireless devices.	Allows students to connect smartphones and laptops to the college network without Ethernet cables.	YouTube, Instagram, Google Classroom
5. Modem	Connects a network to an internet service provider (ISP).	Connects a home network to an ISP through a fiber, cable, or DSL connection, depending on the technology.	Netflix streaming, web browsing
6. Network Interface Card (NIC)	Enables a device to communicate over a network.	Provides an Ethernet or Wi-Fi connection for a laptop or desktop computer.	Zoom meetings, online multiplayer gaming

Task 3:- Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).

1. Access Point (AP): My smartphone connects to the home Wi-Fi network through the wireless access point, which sends and receives data wirelessly.

2. Router: The router forwards my order data from the home network to the internet and directs incoming responses back to my phone.

3. Switch: If a separate network switch is installed, it forwards data between connected devices and the router within the local area network (LAN).

4. Internet and Zomato server: My order request travels across the internet to Zomato's servers, where it is processed and forwarded to the restaurant.

5. Response to my phone: Zomato sends an order confirmation and status updates back through the internet and router to my phone over Wi-Fi.

Task 4:- Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup. Think about devices like switches, powerline adapters, or WiFi extenders.

Priyansh Bhaliya

Two Ways to Connect a Gaming PC to the Internet

Situation: I have a Wi-Fi router, but my gaming PC only has an Ethernet port and is located far away from the router.

1. Wi-Fi Extender with an Ethernet Port



How it works:

- Place the Wi-Fi extender between the router and the gaming PC, where it receives a good Wi-Fi signal.
- The extender connects wirelessly to the router.
- Connect the gaming PC to the extender using an Ethernet cable with RJ-45 connectors.
- The PC can then access the internet through the extender and router.

Advantage: No long Ethernet cable is needed between the router and PC.

2. Powerline Adapter Kit



How it works:

- Connect the first powerline adapter to the router using an Ethernet cable, then plug it into a wall socket.
- Plug the second adapter into a wall socket near the gaming PC.
- Connect the second adapter to the PC using another Ethernet cable.
- The adapters carry network data over the building's electrical wiring.

Advantage: Provides a wired-style connection without running a long Ethernet cable through the house. Actual performance depends on the electrical wiring and circuit layout.

Comparison

Feature	Wi-Fi extender	Powerline adapter
Connection to router	Wireless	Ethernet
Connection to PC	Ethernet	Ethernet
Main requirement	Strong Wi-Fi signal	Suitable electrical wiring
Gaming performance	Can vary with Wi-Fi interference	Can vary with wiring quality and electrical noise